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$NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br) with NLO-Active GaO_4 Tetrahedral Unit: Experimental and ab Initio Studies

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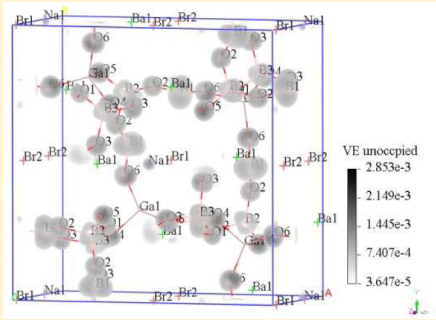
Supporting Information

ABSTRACT: Two gallium borates, $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br), were synthesized by high temperature solution method. The title compounds crystallize in the same space group $P4_2nm$. Their structures feature a 3D $[GaB_4O_9]_\infty$ framework composed of GaO_4 tetrahedra and B_4O_9 groups. $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br) compounds show the largest second harmonic generation responses among alkali metal and alkaline-earth metal gallium borates that are 1.5 and 1.1 times that of KH_2PO_4 (KDP), respectively, and are type I phase-matchable. Combined electronic structure and SHG density calculations as well as local dipole moment analysis have revealed that B−O groups and GaO_4 tetrahedron play major roles in SHG response.



INTRODUCTION

Nonlinear optical (NLO) materials have been catching more and more scientists' attention due to their wide application in laser and photonic technologies.[1−14] To be optically applicable, second harmonic generation (SHG) response and band gap are two fundamental requirements of a ultraviolet (UV) NLO material. In order to design NLO materials with large SHG response, one of the traditional strategies is to introduce NLO-active units, including the parallel π-conjugated units such as BO_3 and B_3O_6,[15−17] the cations with stereochemically active lone pairs,[18−21] such as Pb^{2+} and Bi^{3+} and d^0 transition metal cations with an octahedral coordination geometry,[22−26] such as Ti^{4+}, V^{5+}, and Mo^{6+}. On the other hand, for the obtainment of large band gap, alkali metal and alkaline-earth metal borates were attracting great attention from scientists because of their excellent UV transmittance.

Furthermore, alkali metal and alkaline-earth metal borates were combined with tetrahedral units to generate a number of NLO materials over the past few decades, because the introduction of tetrahedral units can not only enrich the varieties of structure but also cause large distortion of the B−O groups.[27−38] Among them, aluminum borates like BABO ($BaA1_2B_2O_7$) and $BaAlBO_3F_2$ are quite promising as NLO materials, which exhibit relatively large SHG coefficients (>1.5 KDP), short UV absorption edges (≤180 nm), and moderate birefringence (0.063 at 589.3 nm for BABO and 0.052 at 266 nm for $BaAlBO_3F_2$).[28,31,39] Like aluminum borates, gallium borates may be potential NLO materials for their structure units GaO_4 tetrahedron and AlO_4 tetrahedron own similar bond lengths and identical coordination environment. In addition, relative to aluminum borates, gallium borates possess lower reaction temperature and viscosity, which are favorable for crystal growth.

Thus, the combination of B−O group, alkali metal/alkaline-earth metal cations, and GaO_4 tetrahedron results in the discovery of $NaBa_4(GaB_4O_9)_2Br_3$, a new gallium borate halide, and $NaBa_4(GaB_4O_9)_2Cl_3$,[40] whose structure is determined by powder crystal X-ray diffraction previously. $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br) exhibit SHG responses that are 1.5 and 1.1 times that of KDP respectively, which are the largest in alkali metal and alkaline-earth metal gallium borates.[40−44] As mentioned above, gallium borates and aluminum borates own analogous structure units, it is significant to know whether they possess similar properties. Therefore, we synthesized and characterized two aluminum borates, $NaBa_4(AlB_4O_9)_2X_3$ (X = Cl, Br),[38,45] which are isomorphic with title compounds to compare their optical properties. In addition, through our investigation, the great contribution of AlO_4 tetrahedron to SHG response in aluminum borates was reported.[39] While few attention has been paid to the GaO_4 tetrahedron's effect on NLO properties, although several alkali metal and alkaline-earth metal gallium borates with SHG response were discovered.[40−44] Therefore, combined electronic structure and SHG density calculations as well as local dipole moment analysis were performed on $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br) to study the effects of GaO_4 and AlO_4 tetrahedra on optical properties. We believe that our work can not only reveal the relationship between

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structure and property but also benefit other NLO research in gallium borates or aluminum borates.

EXPERIMENTAL SECTION

Reagents. $BaCO_3$ (99.0%), $BaCl_2$ (99.5%), $BaBr_2$ (99.0%), NaCl (99.0%), NaBr (99.0%), Ga_2O_3 (99.9%), Al_2O_3 (99.5%), B_2O_3 (99.99%), LiCl (99.0%), and LiBr (99.5%) were used as received.

Synthesis. Crystals of $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$ were grown from the B_2O_3−LiX (X = Cl, Br) flux system by combining $BaCO_3$, $BaCl_2$, NaCl, Ga_2O_3, B_2O_3, and LiCl with a molar ratio of 3:1:1:1:4:4 for $NaBa_4(GaB_4O_9)_2Cl_3$ and $BaCO_3$, $BaBr_2$, NaBr, Ga_2O_3, B_2O_3, and LiBr with a molar ratio of 3:1:1:1:4:4 for $NaBa_4(GaB_4O_9)_2Br_3$. The mixtures were put into platinum crucibles, gradually heated up to 900 °C and held at this temperature for 12 h. Then, they were cooled to 700 °C at a rate of 4 °C/h and finally quenched to room temperature. Finally, colorless bulk crystals of $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$ were obtained and separated for further single crystal X-ray study.

Polycrystalline samples of $NaBa_4(GaB_4O_9)_2Cl_3$, $NaBa_4(GaB_4O_9)_2Br_3$, $NaBa_4(AlB_4O_9)_2Cl_3$, and $NaBa_4(AlB_4O_9)_2Br_3$ were prepared by standard solid state reaction. Stoichiometric proportions of raw materials were mixed and presintered at 500 °C for 24 h. Then the mixtures were elevated to sintering temperatures of 700, 740, 720, and 750 °C, respectively, and held at these temperatures for 1 week with several intermittent grindings. The powder samples were characterized by powder X-ray diffraction.

Structure Determination. Single-crystal X-ray diffraction data were collected with the use of monochromatic Mo Kα radiation at 293(2) K on an APEX II CCD diffractometer and integrated with the SAINT program.[46] The numerical absorption corrections were carried out using the SCALE program for area detector. All calculations were performed with programs from the SHELXTL crystallographic software package.[47] The structure was checked for missing symmetry elements with PLATON.[48] Final least-squares refinement on F_o^2 with data having $F_o^2 \geq 2\sigma(F_o^2)$ includes an isotropic displacement parameters for all atoms. Details of crystal parameters, data collection, and structure refinements are given in Table 1. The atomic coordinates and the equivalent isotropic displacement parameters are summarized in Table S1 in the Supporting Information. The thermal factors of Na(1), Cl(1), and Br(2) are large compared with other atoms, which indicate the possibility of their site deficiencies.[40]

Powder X-ray Diffraction. Powder X-ray diffraction (XRD) data were collected using a Bruker D2 PHASER diffractometer using Cu Kα radiation. The 2θ range was 10°−70°, with a scan step width of 0.02° and a counting time of 1 s/step. The experimental powder XRD data are in good agreement with the calculated pattern on the basis of the single-crystal crystallographic data of $NaBa_4(AB_4O_9)_2X_3$ (A = Al, Ga; X = Cl, Br; Figure S1 in the Supporting Information).

Infrared and UV−vis−NIR Diffuse Reflectance Spectroscopy. The sample was mixed thoroughly with dried KBr (2 mg samples mixed with 200 mg KBr). Infrared spectrum ranged from 400 to 4000 cm^{-1} was recorded on Shimadzu IR Affinity-1 Fourier transform infrared spectrometer.

The transmission spectra were measured on a Shimadzu SolidSpec-3700DUV UV−vis−IR spectrophotometer with a range from 190 to 2600 nm. Polytetrafluoroethylene is used as a standard. Reflectance spectra were converted to absorbance using the Kubelka−Munk function.[49,50]

Table 1. Crystal Data and Structure Refinements for $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$

empirical formula	$NaBa_4(GaB_4O_9)_2Cl_3$	$NaBa_4(GaB_4O_9)_2Br_3$
temperature	296(2) K	
wavelength	0.71073 Å	
crystal system	tetragonal	
space group	$P4_2nm$	
formula weight	1192.62	1326.00
a (Å)	12.1033(10)	12.347(17)
b (Å)	12.1033(10)	12.347(17)
c (Å)	6.8329(12)	6.911(19)
Z, Volume (Å^3)	2, 1001.0(2)	2, 1053(3) (11)
ρ_{Calcd} (Mg/m^3)	3.957	4.180
μ(/mm)	10.907	15.673
F(000)	1064	1172
R(int)	0.0462	0.0651
Goodness-of-fit on F^2	1.079	1.137
Final R indices [$F_o^2 > 2\sigma(F_o^2)$][a]	R1 = 0.0286, wR2 = 0.0671	R1 = 0.0335, wR2 = 0.0951
R indices (all data)[a]	R1 = 0.0324, wR2 = 0.0690	R1 = 0.0384, wR2 = 0.0980
Extinction coefficient	0.0051(3)	0.0047(4)
Absolute structure parameter	0.00(4)	0.02(5)
Largest diff. peak and hole (e·Å^{-3})	1.220 and −1.393	1.166 and −2.211

[a] $R_1 = \Sigma||F_o| - |F_c||/\Sigma|F_o|$ and $wR_2 = [\Sigma w(F_o^2 - F_c^2)^2/\Sigma w F_o^4]^{1/2}$ for $F_o^2 > 2\sigma(F_o^2)$

Thermal Analysis. The thermal gravity-differential scanning calorimetric (TG-DSC) analysis was investigated using a NETZSCH STA 449C simultaneous thermalanalyzer under flowing nitrogen gas. The samples and reference Al_2O_3 were elevated at a rate of 10 °C/min in the range of 25−1200 °C.

Second-Harmonic Generation Effects. Powder second-harmonic generation (SHG) measurements of $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br) were performed on a modified Kurtz−Perry[51] system using a pulsed Nd:YAG laser with a wavelength of 1064 nm. SHG efficiency has been shown to depend strongly on particle size, and thus, polycrystalline samples were ground and sieved into distinct particle size ranges (<20, 20−38, 38−55, 55−88, 88−105, 105−150, and 150−200 μm). In order to make relevant comparisons with known SHG materials, crystalline KDP served as the standard.

COMPUTATIONAL METHODS

To better understand the origin of the enhanced SHG response, the electronic structure and the optical properties of a series of borates were calculated using density functional theory[52,53] as implemented in the CASTEP package.[54] During the calculation, the generalized gradient approximation (GGA) with Perdew−Burke−Ernzerhof (PBE) functional[55,56] and the norm-conserving pseudopotential (NCP)[57,58] were adopted, the following orbital electrons were treated as valence electrons: B:$2s^22p^1$, O:$2s^22p^4$, Cl:$3s^23p^5$, Br: $4s^24p^5$, Al: $3s^23p^1$, Ga:$3d^{10}4s^24p^1$,Ba:$5s^25p^66s^2$ and Na: $2s^22p^63s^1$. The kinetic energy cutoffs of 800 eV for $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br) were chosen, and the numerical integration of the Brillouin zone was performed using a 3 × 3 × 6 for $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br) Monkhorst−Pack k-point sampling, respectively. The other calculation

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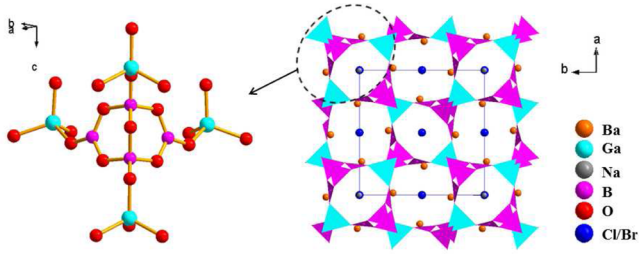

Figure 1. Structure of $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br).

parameters and convergent criteria were the default values of the CASTEP code.

The imaginary part of the dielectric function $\varepsilon_2(\omega)$ can be calculated from the electronic transition between the occupied and unoccupied states based on the obtained electronic structures, which is given by

$$\varepsilon_2(q \to O_{\hat{u}}, h\omega) = \frac{2e^2\pi}{\Omega\varepsilon_0}\sum_{kcv}|\langle\varphi_k^c|u\cdot r|\varphi_k^v\rangle|^2\delta[E_k^c - E_k^v - E] \quad (1)$$

In which, Ω is the volume of the elemental cell, v, and c represents the valence and conduction bands, respectively, and u is the vector defining the polarization of the electric field of the incident light. Since the dielectric constant describes a causal response, the real and imaginary parts are linked by a Kramers−Kronig transform. This transform is used to obtain the real part of the dielectric function, $\varepsilon_1(\omega)$, and then the refractive index n.

Using the length-gauge formalism at a zero frequency limit,[59] the SHG coefficients can be estimated, which can be described as a later developed transform by Zhang et al.:

$$\chi_{\alpha\beta\gamma}^{(2)} = \chi_{\alpha\beta\gamma}^{(2)}(\mathrm{VE}) + \chi_{\alpha\beta\gamma}^{(2)}(\mathrm{VH}) \quad (2)$$

Here, the total SHG coefficient $\chi_{\alpha\beta\gamma}^{(2)}$ is divided into the contribution from two different processes which are virtual-electron (VE) and virtual-hole (VH).[60] The theoretical methods have been applied with success to analyze the SHG coefficients of NLO crystal materials in previous study.[61,62]

RESULTS AND DISCUSSION

Structure. $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$ are isostructural and crystallize in the polar space group $P4_2nm$ (No. 102). Their structures feature a 3D $[GaB_4O_9]_\infty$ framework composed of GaO_4 tetrahedra and B_4O_9 groups (Figure 2). The B_4O_9 groups are built up of two BO_3 triangles

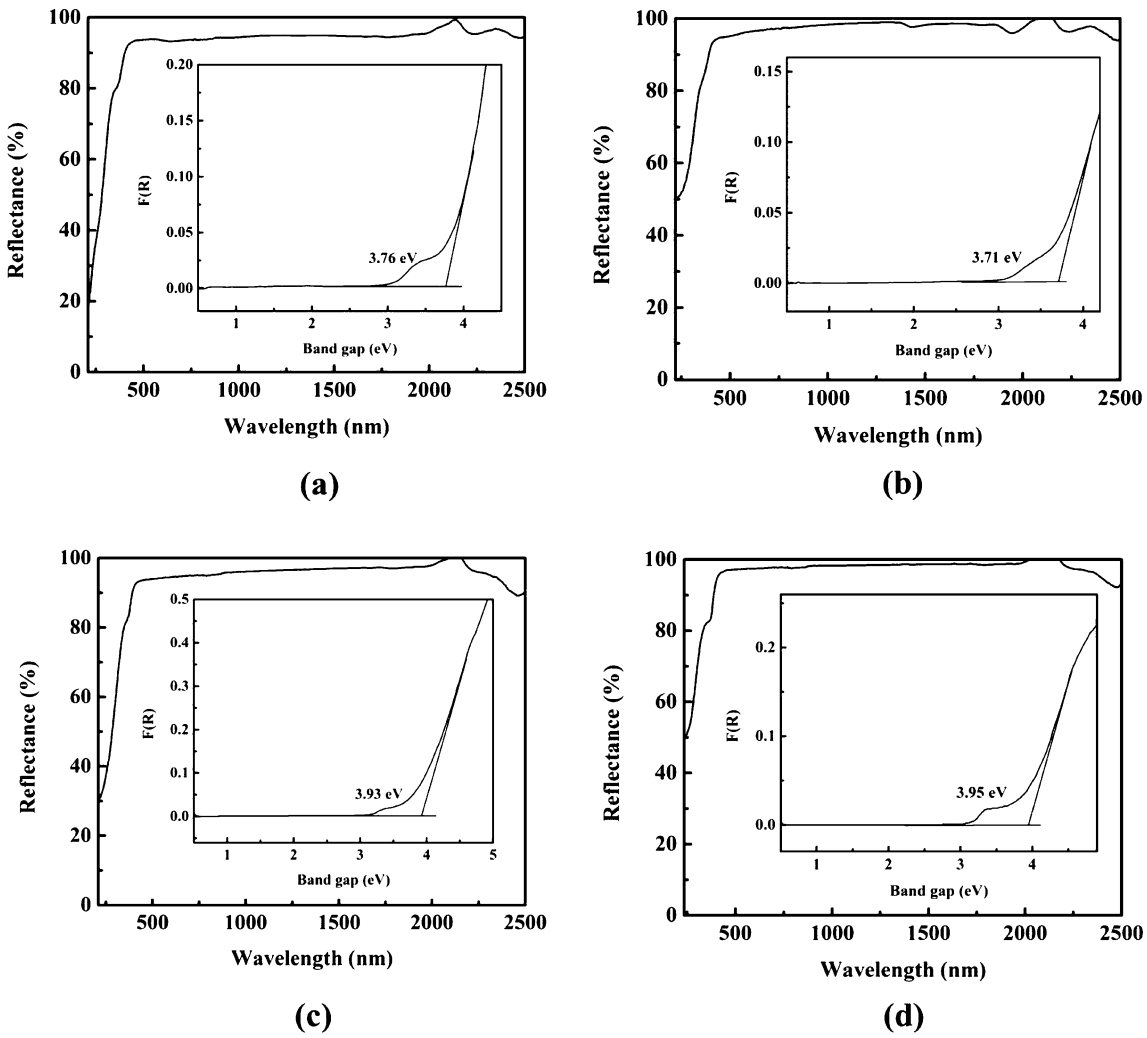

Figure 2. UV−VIS−NIR diffuse reflectance spectra of $NaBa_4(GaB_4O_9)_2Cl_3$ (a), $NaBa_4(GaB_4O_9)_2Br_3$ (b), $NaBa_4(AlB_4O_9)_2Cl_3$ (c), and $NaBa_4(AlB_4O_9)_2Br_3$ (d).

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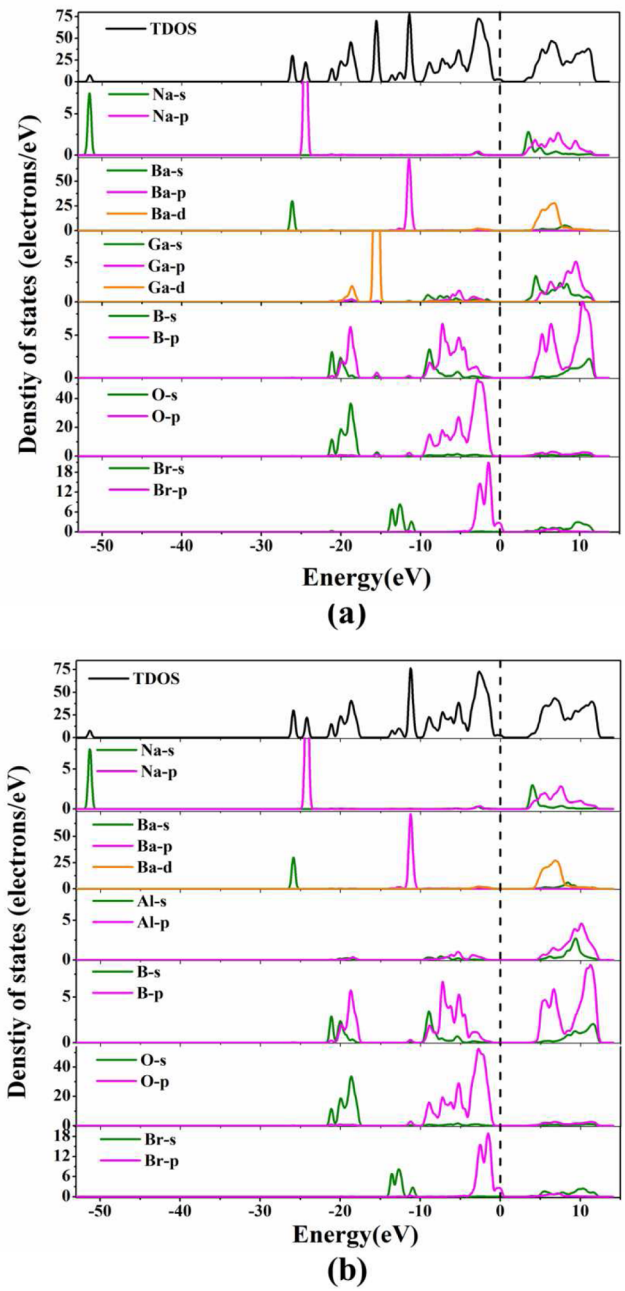

Figure 3. DOS of (a) $NaBa_4(GaB_4O_9)_2Br_3$ and (b) $NaBa_4(AlB_4O_9)_2Br_3$.

and two BO_4 tetrahedra and each of them is connected by four GaO_4 tetrahedra to form the 3D network. While the Na, Ba and Cl or Br atoms are filled in the space created by the network (Figure 1). In $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$, the B^{3+} cations are in 3-fold triangular or 4-fold tetrahedral coordination environments with bond distances in the range 1.361(16)-1.396(16) Å and 1.409(15)-1.54(3) Å respectively. Bond valences sums (BVS) calculations for B are 3.03 for B(1), 3.18 for B(2) and 3.03 for B(3) in $NaBa_4(GaB_4O_9)_2Cl_3$ and 2.86 for B(1), 2.90 for B(2) and 2.89 for B(3) in $NaBa_4(GaB_4O_9)_2Br_3$, which confirms that the B atom in the compounds adopt normal valence state. The Ga^{3+} cations are in 4-fold tetrahedral coordination environments with bond distances in the range 1.82(6)-1.883(11) Å (Table S2 in the Supporting Information). The Ga^{3+} cations are slightly

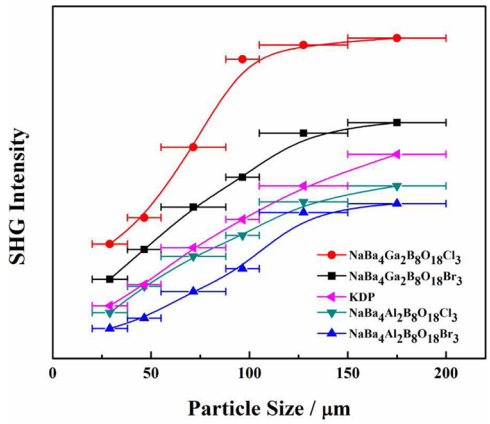

Figure 4. Phase-matching curves of $NaBa_4(AlB_4O_9)_2Cl_3$, $NaBa_4(AlB_4O_9)_2Br_3$, $NaBa_4(GaB_4O_9)_2Cl_3$, and $NaBa_4(GaB_4O_9)_2Br_3$.

displaced toward one face of their oxide tetrahedron, resulting in three "short" and one "long" Ga−O bonds [1.820(6), 1.823(5), 1.823(5), and 1.840(6) Å for $NaBa_4(GaB_4O_9)_2Cl_3$ and 1.830(12), 1.842(9), 1.842(9), 1.883(11) Å for $NaBa_4(GaB_4O_9)_2Br_3$]. BVS calculation for Ga is 3.08 for $NaBa_4(GaB_4O_9)_2Cl_3$ and 2.84 for $NaBa_4(GaB_4O_9)_2Br_3$, which further prove that the coordinations of Ga atoms are very reasonable. In addition, the bond valences of the O anions are closed to each other with an average value of 2.07 for $NaBa_4(GaB_4O_9)_2Cl_3$ and 1.89 for $NaBa_4(GaB_4O_9)_2Br_3$.

Thermal Analysis. The TG-DSC curves of $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br) were performed. As shown in Figure S2 in the Supporting Information, the only one endothermic peak of $NaBa_4(GaB_4O_9)_2Cl_3$, $NaBa_4(GaB_4O_9)_2Br_3$ $NaBa_4(AlB_4O_9)_2Cl_3$ and $NaBa_4(AlB_4O_9)_2Br_3$ locates at 950, 933, 962, and 947 °C respectively with obvious weight loss. And evidence of the fusion of the samples were found after the measurements. These indicate that $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$ melt prior to $NaBa_4(AlB_4O_9)_2Cl_3$ and $NaBa_4(AlB_4O_9)_2Br_3$, respectively, in crystal grow process.

Infrared Spectrum. The IR spectra of $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$ are depicted in Figure S3 in the Supporting Information. The assignments of the peaks are shown in Table S3 in the Supporting Information. The peaks observed in these IR spectra are well in accordance with other compounds containing BO_3, BO_4, and GaO_4 groups.[38,44]

UV−vis−NIR Diffuse Reflectance Spectroscopy and Electronic Structure Calculations. The UV−vis−IR diffuse reflectance spectra of $NaBa_4(AlB_4O_9)_2Cl_3$, $NaBa_4(AlB_4O_9)_2Br_3$, $NaBa_4(GaB_4O_9)_2Cl_3$, and $NaBa_4(GaB_4O_9)_2Br_3$ are measured indicating their band gaps 3.93, 3.95, 3.76, and 3.71 eV respectively (Figure 2). From $NaBa_4(AlB_4O_9)_2X_3$ (X = Cl, Br) to $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br), there is a small red shift in the spectra, which will be further discussed through electronic structure calculations.

The first-principles electronic structures of $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br) were obtained to further elucidate the microscopic origins of the optical properties. The band structures of $NaBa_4Ga_2B_8O_{18}Br_3$, $NaBa_4Ga_2B_8O_{18}Cl_3$, $NaBa_4Al_2B_8O_{18}Br_3$, and $NaBa_4Al_2B_8O_{18}Cl_3$ are shown in Figure S4 in the Supporting Information. All of these materials are direct band gap compounds with the band gap of 3.75 [for $NaBa_4(GaB_4O_9)_2Cl_3$], 2.98 [for $NaBa_4(GaB_4O_9)_2Br_3$], 3.90 [for $NaBa_4(AlB_4O_9)_2Cl_3$], and 3.43 eV [for $NaBa_4(AlB_4O_9)_2Br_3$], respectively, which are

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Table 2. Magnitude of the Dipole Moments of the BO_3 Triangle, BO_4 Tetrahedra, and AlO_4 or GaO_4 Tetrahedra along *c* Axis in the Unit Cell and the SHG Intensity

magnitude (debye)	$NaBa_4(GaB_4O_9)_2Cl_3$	$NaBa_4(AlB_4O_9)_2Cl_3$	$NaBa_4(GaB_4O_9)_2Br_3$	$NaBa_4(AlB_4O_9)_2Br_3$
BO_3	8.35	−2.86	7.12	4.87
BO_4	0.28	1.55	−6.26	−5.95
B−O groups	8.63	−1.31	0.86	−1.08
GaO_4/AlO_4	4.67	−2.19	3.99	2.00
total of anion groups	13.30	−3.50	4.85	0.92
ratio of GaO_4/ AlO_4 to total	0.35	0.63	0.82	0.65
SHG intensity (KDP)	1.5	0.9	1.1	0.8

Table 3. Calculated SHG Tensors and Experimental SHG Intensities of $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br)[a]

compound	calculated SHG tensors (pm/V)	SHG intensity
$NaBa_4(GaB_4O_9)_2Br_3$	$d_{31} = d_{15} = 0.50$	1.1KDP
$NaBa_4(GaB_4O_9)_2Cl_3$	$d_{31} = d_{15} = 0.70$	1.5KDP
$NaBa_4(AlB_4O_9)_2Br_3$	$d_{31} = d_{15} = 0.20$	0.8KDP
$NaBa_4(AlB_4O_9)_2Cl_3$	$d_{31} = d_{15} = -0.45$	0.9KDP

[a]The SHG tensor of KDP is d_{36} (KDP) = 0.39 pm/V.

smaller than the experimental values (3.76, 3.71, 3.93, and 3.95 eV, respectively). The underestimated band gap is generally due to insufficient accuracy of the exchange correlation energy calculated.[63]

To better understand the relationship between electronic structures and optical properties, we compare the density of states (DOS) of $NaBa_4(GaB_4O_9)_2Br_3$ and $NaBa_4(AlB_4O_9)_2Br_3$ (shown in Figure 3). At the valence band (VB), the two compounds have similar electronic structures: the inner valence orbitals for the constituent ions (i.e., 2s orbitals for oxygen, 2s and 2p orbitals for boron and sodium, and 4s and 4p orbitals for barium and bromine) are located at the deep region of the valence bands below −10 eV. The valence bands from −10 to 0 eV are mainly composed of the 2p orbitals of oxygen and boron as well as 4p orbitals of bromine, while the O 2p orbitals have dominant contribution to the top of VB. The large hybridization between oxygen and boron orbitals indicates the formation of strong covalent bonds in the B−O groups. Moreover, the Ga 4s and 4p orbitals or Al 3p orbital also have some contribution to the upper parts of the VB, and their hybridization with O 2p orbitals is clearly exhibited. While in the conduction band (CB) region, for $NaBa_4(AlB_4O_9)_2Br_3$, the conduction band minimum (CBM) is dominated by B 2p, Ba 5d, and Na 2s orbitals. However, in $NaBa_4(GaB_4O_9)_2Br_3$, besides B 2p, Ba 5d, and Na 2s orbitals, Ga 4s orbital also occupy the CBM and located nearer to VBM than B 2p orbital, which cause the small red shift of the band gaps from $NaBa_4(AlB_4O_9)_2Br_3$ to $NaBa_4(GaB_4O_9)_2Br_3$. In addition, the DOS of $NaBa_4(AlB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Cl_3$ (shown in Figure S5 in the Supporting Information) is similar to that of $NaBa_4(AlB_4O_9)_2Br_3$ and $NaBa_4(GaB_4O_9)_2Br_3$, respectively. Since the optical properties of a crystal in the visible and UV regions are mainly determined by the electronic states close to the energy band gap,[64] it is expected that the B, O, Ga, and Na ions have major contribution to the SHG response in $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$.

Second-Harmonic Generation Effects. SHG measurements reveal that $NaBa_4(AlB_4O_9)_2Cl_3$, and $NaBa_4(AlB_4O_9)_2Br_3$ exhibit a SHG response of 0.9 and 0.8 times that of KDP respectively, while the SHG intensity of $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$ are larger (1.5 and 1.1 times of KDP respectively) shown in Figure 4. The plots of particle size versus SHG efficiency indicate that all the four compounds are type I phase-matchable.

In order to investigate the origin of the SHG response of the four compounds, local dipole moment analysis as well as SHG coefficient calculations were performed.

(A). Local Dipole Moment Analysis. We computed the local dipole moments of the anion groups using a bond-valence approach.[65,66] As is shown in Table 2, only the dipole moments of *c* direction in the unit cell are depicted because the dipole moments of other direction are zero in the unit cell according to the symmetrical characteristic because the four compounds all crystallize in the tetragonal space group $P4_2nm$ and own a 4_2 screw axis along *c* axis.

The anion groups' total dipole moments of $NaBa_4(AlB_4O_9)_2Br_3$, $NaBa_4(AlB_4O_9)_2Cl_3$, $NaBa_4(GaB_4O_9)_2Br_3$, and $NaBa_4(GaB_4O_9)_2Cl_3$ are 0.92, 3.50, 4.85, and 13.30 debye respectively, which indicate that the SHG response (0.8, 0.9, 1.1, and 1.5 times that of KDP, respectively) increase with their dipole moments in the four compounds. To discuss the contribution of GaO_4 or AlO_4 tetrahedra to the SHG intensity, the dipole moment's ratios of GaO_4/AlO_4 to the total of anion groups are considered. In the four compounds, all the ratios are more than 60 percent except $NaBa_4(GaB_4O_9)_2Cl_3$'s (0.35), which indicates the GaO_4/ AlO_4 tetrahedra's great contribution to SHG response.

(B). SHG Coefficient Calculations. The calculated SHG coefficients of title compounds are shown in Table 3 along with the experimental values. The good consistency between the calculated values and experimental SHG intensities demonstrates the validity and accuracy of the first-principles method on the studied compounds.

To give an intuitive visualization of the orbitals that contribute to the SHG effects in the real space, the SHG-densities of occupied and unoccupied states in $NaBa_4(GaB_4O_9)_2Br_3$ and $NaBa_4(AlB_4O_9)_2Br_3$ are performed. The SHG densities of occupied states (see Figure 5a,b) show that the SHG densities on O atoms are dominant. For the unoccupied states (Figures 5c,d) the SHG densities mainly accumulated on O and B atoms, which proves an significant role of B−O groups. In addition, four O atoms in GaO_4 tetrahedron all exhibit large SHG densities (shown in Figures 5a,c), which indicates that the GaO_4 tetrahedra may also play important roles in the SHG process. Similarly, AlO_4 tetrahedra may make significant contribution to the SHG response because of the large SHG densities of their O atoms (shown in Figures 5b,d). The SHG densities of $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(AlB_4O_9)_2Cl_3$ are analogous with $NaBa_4(GaB_4O_9)_2Br_3$ and $NaBa_4(AlB_4O_9)_2Br_3$ respectively, depicted in Figure S6 in the Supporting Information.

Based on DOS, local dipole moment and SHG density analyses, it is indicated that B−O groups, GaO_4 tetrahedron,

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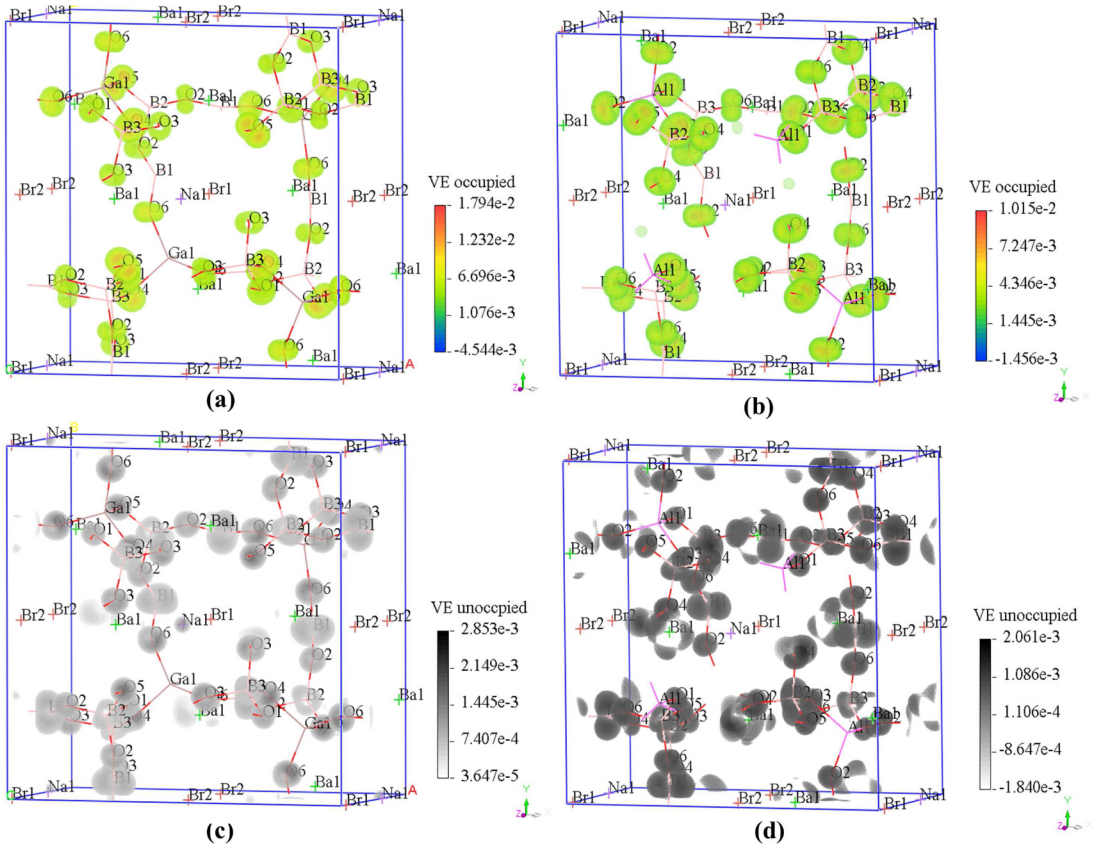

Figure 5. Occupied SHG densities of (a) $NaBa_4(GaB_4O_9)_2Br_3$ and (b) $NaBa_4(AlB_4O_9)_2Br_3$. The unoccupied SHG densities of (c) $NaBa_4(GaB_4O_9)_2Br_3$ and (d) $NaBa_4(AlB_4O_9)_2Br_3$.

and AlO_4 tetrahedron play important roles in SHG response. GaO_4 tetrahedron may be a NLO-active unit.

■ CONCLUSIONS

The combination of GaO_4 tetrahedron and B−O group generates two gallium borate halides, $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br). They feature a 3D $[GaB_4O_9]_\infty$ framework structure, which consists of GaO_4 tetrahedra and B_4O_9 groups. $NaBa_4(GaB_4O_9)_2Cl_3$ and $NaBa_4(GaB_4O_9)_2Br_3$ show the largest SHG responses (1.5 and 1.1 times that of KDP, respectively) among alkali metal and alkaline-earth metal gallium borates, which are larger than that of $NaBa_4(AlB_4O_9)_2Cl_3$ and $NaBa_4(AlB_4O_9)_2Br_3$ (0.9 and 0.8 times that of KDP, respectively). The B−O groups, GaO_4 tetrahedron, and AlO_4 tetrahedron have significant contribution to SHG response and GaO_4 tetrahedron may become a NLO-active unit. The band gaps of the title compounds are smaller than that of $NaBa_4(AlB_4O_9)_2X_3$ (X = Cl, Br), which is due to Ga−4s state's occupation of the CBM illuminated by DOS analysis. We believe that these results are essential for the elucidation of the structure−property relationship and can give important guidance on further NLO study in gallium borates or aluminum borates.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.6b00265.

Atomic coordinates and equivalent isotropic displacement parameters, selected bond distances and angles and IR spectra of $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br), experimental and calculated XRD patterns, TG-DSC curves and band structures of $NaBa_4(AB_4O_9)_2X_3$ (A = Ga, Al; X = Cl, Br), SHG densities and DOS of $NaBa_4(AB_4O_9)_2Cl_3$ (A = Ga, Al), X-ray crystallographic files, in cif format, for $NaBa_4(GaB_4O_9)_2X_3$ (X = Cl, Br). (PDF)

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Notes

The authors declare no competing financial interest.

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